

Section 6.4: Internet Developments and Issues

In **1993 the web browser Mosaic was launched**. Mosaic is credited with making the **Internet easily available to the general public**.

Ever since then, the Internet has **undergone constant change and development**. This section looks at some of these developments including **Web 2.0, blog, wikis, digital media uploading websites** and **social networking websites**.

Key Concepts of this section:

- # Understand the developments that came with the second generation of the Internet (**Web 2.0**).
- # Know the uses of **blog's, wikis, digital media websites** and **social networking**.
- # Be able to discuss the **security issues of data transfer over the Internet**.

Web 2.0

- # Web 2.0 is the term given to the **second generation of the Internet**.

The **developments made to Web 2.0** focus on giving people the **ability to share information online**. Some of these developments include:

- **Blog's**
- **Wikis**
- **Digital media uploading websites**
- **Social networking**



Key Words:

Web 2.0, Blog's, Wikis,
Digital Media, Social
Networking

Uses of Blog's, wikis, digital media websites and social networks

1 Blog's (web logs)

A blog is a kind of **online journal** or a **log** where '**bloggers**' write about **topics that are important to them**.

Blog's can be about anything that the blogger wants to share with other Internet users and interesting Blog's can attract **millions of visitors**.

Some current popular Blog's include:

- [FAILBlog](#)
- [Mashable](#)
- [Engadget](#)
- [Huffington Post](#)
- [TMZ](#)



2 Wikis

Wikis are **interactive web pages** that can be **viewed and edited by users**.

Anyone with a web browser can visit a wiki and make **changes to the content**. This has both advantages and disadvantages as shown below:

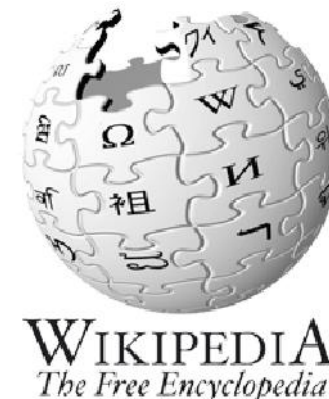
Advantage

Information is quickly added as many people collaborate their knowledge.

Disadvantage

One of the problems with letting everyone add to the wiki is that sometimes incorrect information is posted.

The best known wiki on the web is probably [Wikipedia](#).



3 Digital media sharing websites

These websites allow users to **upload video and images** which can then be **viewed by other people**.

Video sharing websites, in particular, are helping to **change the way in which we can learn on the Internet**. For example, YouTube has a huge amount of '**how to**' and '**tutorial**' videos for all sorts of topics.

One of the advantages of digital media websites is that the owners of the videos and images do not need to waste their own hard drive space storing them. The files are held '**remotely**' on the websites servers.

The best known video sharing website is [YouTube](#) and for images [Flickr](#).



4 Social networking websites

These websites create '**communities**' of **people** who can share information about their **lives, interests** and **hobbies** with each other.

They also make it easy for people to **make new friends** and to **keep in touch with each other** no matter where they are in the world.

Common features of social networking websites include:

- **Status updates** - letting friends know what you have been up to
- **Message boards** - where groups of friends can post each other messages
- **Private email** - this allows people to send private messages
- **Video conferencing** - where you can have face-to-face chats using web cams.

The best know social networking websites are [Facebook](#), [Twitter](#) and [MySpace](#).



Issues with information on the Internet

- # Although the Internet has many many uses there are some issues that can cause us problems if we are not careful.

Some of these issues include:

- **Unreliable information**
- **Undesirable information**



Key Words:

Unreliable, Undesirable

Description of issues with the Internet regarding information

1 Unreliable information

Just because something is written on the Internet does not necessarily make it **true or correct**. **Anyone can make a website** and a lot of the information held on these websites is **never checked or verified**.

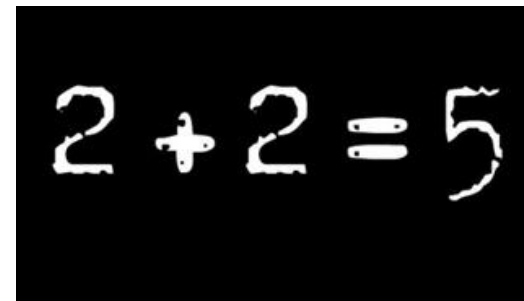
Wikis (like Wikipedia) allow anyone to change the information held on the site. This means that sometimes what is written is completely incorrect.

In fact, some websites **deliberately set out to fill their pages with incorrect information** just to cause a nuisance to people. Examples [here](#) and [here](#).



Ways to check that online information is correct

1. Use **well known websites** that you have heard of (bba, yahoo news etc).
2. Use **official websites** (band websites, football clubs etc).
3. **Verify the information by checking it on several different websites.** If each one says the same thing you can be sure its correct.
4. Try and **stay away from sites where just anyone can post the information** (like ask.com or answers.com).
5. Use websites that are **updated regularly**. Fake websites are not usually kept up to date as much.



NOTE!

The Internet is still an amazing tool for research. You just have to know how to use it!!!!

2 Undesirable information

Undesirable websites are those that include **content that is offensive to individuals or even races**. Undesirable content can include:

- **Pornography**
- **Violent videos or images**
- **Content promoting hatred**
- **Racial content**

We **cannot fully censor (block)** this type of content from the Internet as many of the sites are based in **Countries that do not have laws against it**. However you can take steps to reduce the chance of you accidentally stumbling across undesirable information:





Ways to avoid undesirable information

1. Stick to **well known websites** that you have heard of.
2. Use **built-in search engine filters**. Google has a feature which will exclude search results that contain undesirable content.
3. Use **parental controls** which are built into most modern browsers. These block access to many undesirable websites.

Security of information on the Internet



Key Words:

Information Security,
Phishing, Pharming,
Spam

We store a lot of information on our computers and mobile devices. Some of this information is **sensitive and personal** and we would not want it falling into the wrong hands. Sensitive information can include:

- **Credit card details**
- **Banking information**
- **Addresses / Phone numbers of friends and family**

As soon as we log onto the Internet our sensitive information is at risk of '**hackers**' and '**fraudsters**' from **trying to steal it**. Common methods include:

- **Phishing**
- **Pharming**
- **Spam**

How do hackers and fraudsters try to steal our data?

1 Phishing

'**Phishing**' is used to describe methods that **thieves use to 'fish' for our data**.

This is where fraudsters try and **trick us into giving up our bank details, credit card details, usernames and passwords**.

Phishing is usually **carried out over emails pretending to be from legitimate organisations** like banks and building societies.

The emails usually say that there is a problem with your account and then they ask you to provide your usernames, passwords or account numbers so that the problem can be rectified.

If you provide these details then the criminals will be able to use them to help themselves to your cash.



Ways to reduce the risk of getting 'phished'

1. Never ever give your **bank details or passwords out over email**. Banks **NEVER** ask their customers for this information.
2. Some phishing scams promise you enormous wealth "**If you pay a small administration fee**". If something sounds too good to be true then it probably is and you would be best off ignoring the mail.
3. Report any phishing attempts to your email account provider.
4. Don't respond to emails from people you do not know.



NOTE!

In **2007**, phishing cost the USA alone around **3.7 BILLION dollars**.

Phishing is extremely profitable to thieves but they **rely on our stupidity** to hand over our sensitive data.

2 Pharming

The intention of pharming is the same as phishing..... to obtain personal information such as **usernames, passwords** and **bank details** etc. Pharming usually targets users of online banking or shopping websites.

The way this is done however is slightly different. '**Pharmers**' infect **legitimate websites with malicious code** that will '**re-direct**' you to their **bogus version of the website**.

The bogus website will look very similar, or even identical, as the legitimate website. This makes pharming very dangerous and difficult to detect.



Ways to reduce the risk of getting 'pharmed'

1. Check the **url (web address)** of the website before you enter personal information. The **bogus website** will usually have a **slightly different**



address to the **legitimate website**.

2. Make sure that you are on a **secure website** before entering personal information. The way to tell is to look at the url:

- A secure connection's address will begin with **https**
- An unsecure connection will begin with **http**

3 Spam

Spam is '**junk email**' that is **sent automatically** to **thousands, or even millions, of email inboxes**. Spam over-runs your inbox very quickly and it is a pain to have to delete all of these unwanted emails every day.

'**Spammers**' send these unwanted emails for a range of different reasons including:

- **Advertising low quality products**
- **Spreading viruses**
- **Phishing purposes**

Spammers get your email address from a range of sources. This is called '**harvesting email addresses**'. The sources include:

- **Mailing lists**
- **Chat rooms**
- **Programs that scour websites for email addresses**
- **Bought illegally from dishonest employee's of internet companies**



Ways to reduce the risk of spam

1. **Spam filters** are included with most email accounts. These can be used to **automatically separate junk email from legitimate emails**. You can then quickly delete the contents of the junk email folder.
2. **Don't post your email address in public websites** like chat rooms or social networking sites.



NOTE!

In **2011** it is estimated that **seven trillion spam emails** were sent over the Internet (that's seven thousand billion).

Spam emails account for around **80% of all the emails** sent around the world.

Activity!

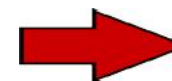
[Internet Developments - Questions](#)

Click the above task and follow the instructions given on the sheet. Make sure that you complete all questions in full.



Previous - Capabilities and Limitations of ICT

Next - Health and Safety



Links to Theory Units:

[Section 1: Types and Components of Computer Systems](#)

[Section 4: Computer Networks](#)

[Section 7: The Ways in which ICT is used](#)

[Section 2: Input and Output Devices](#)

[Section 5: Data Types](#)

[Section 7: Part 2](#)

[Section 3: Storage Devices and Media](#)

[Section 6: The Effects of Using ICT](#)

[Section 8: Systems Analysis and Design](#)

Links to Practical Units:

[Section 9: Communication](#)

[Section 12: Integration](#)

[Section 15: Web site Authoring](#)

[Section 10: Document Production](#)

[Section 13: Output Data](#)

[Section 16: Presentation Authoring](#)

[Section 11: Data Manipulation](#)

[Section 14: Data Analysis](#)

Back to top